INCH-POUND

MIL-DTL-32752
18 September 2023

DETAIL SPECIFICATION

CLOTH, WATERPROOF, STRETCH, MOISTURE VAPOR PERMEABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of waterproof, breathable, laminated stretch cloth used in the manufacture of anti-exposure socks.

1.2 Classification.

1.2.1 Type. The cloth is furnished in one type (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommendation for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all the specified requirements of documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the exact specific herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop B120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocCom.fct@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

(Copies of this document are available online at https://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-3064	Evaluation of Quality of Textile Material

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawing, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP 1	Gray Scale for Color Change
AATCC EP 2	Gray Scale for Staining
AATCC EP 9	Visual Assessment of Color Difference of Textiles
AATCC TM8	Colorfastness to Crocking: Crockmeter
AATCC TM15	Colorfastness to Perspiration
AATCC TM22	Water Repellency: Spray
AATCC TM116	Colorfastness to Crocking: Rotary Vertical Crockmeter
AATCC TM127	Water Resistance: Hydrostatic Pressure
AATCC TM135	Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQ)

ASQ/ANSI Z1.4	Procedures, Sampling and Tables for Inspection by Attributes

(Copies of this document are available from https://www.asq.org.)

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ASTM INTERNATIONAL

ASTM D2594/D2594M	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3787	Standard Test Method for Bursting Strength of Textiles – Constant-Rate-of-Traverse (CRT) Ball Burst Test
ASTM D3776/D3776M	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM E96/E96M	Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. The sock shall match the standard sample for shade and appearance on the face side and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.3 Cloth. The cloth shall be one type, laminate cloth consisting of a jersey knit nylon/spandex face and a jersey knit nylon backer. Unless otherwise specified in the contract (see 6.2), the color shall be sage green #3735 on the face and white on the back matching the standard sample. The finished cloth shall meet the requirements specified in table I and 3.4 through 3.8 when tested, as specified in 4.6.

3.4 Width. For Government procurements only, unless otherwise specified, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.5 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished full width in continuous lengths, each not less than 20 yards.

3.6 Face identification. The face side of the solid shade dyed cloth shall be identified by stamping that side with the word “FACE” at each end of the roll.

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Table I. Physical requirements.

Characteristic	Minimum
Ball Burst Test (mm)	No deformation ≥ 70
Weight (ounces/yard^2)	8.50±.5
Moisture Vapor Transmission Rate - Inverted	≥ 4000
Stretch % % Elongation Warp % Recovery Warp	 ≥30 ≥90
Colorfastness after Crocking Dry Wet	 ≥3-4 ≥3-4
Low Pressure Hydrostatic Testing (0.7psi for 3 minutes) - Initial - After 20 Home Laundry Cycles - After Synthetic Perspiration - After Aircraft Fluid	 No leakage or delamination No leakage or delamination No leakage or delamination No leakage or delamination
Colorfastness to Perspiration Shade Change (nylon)	≥3-4
Spray Rating	≥ 90, 90, 90
Dimensional Stability (%)	-8.00 ≤ result ≤ 8.00 -8.00 ≤ result ≤ 8.00

3.7 Roll identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.8 Workmanship. The finished cloth shall be uniform in quality and free from irregular defects that can adversely affect usage or durability and the defects specified in 4.4.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

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4.2 First article inspection. The first article, submitted in accordance with 3.1, shall be inspected as specified in 4.3.2 through 4.4 for compliance with design, construction, workmanship and dimensional requirements.

4.3 Conformance inspection. Conformance testing shall consist of all examinations and tests specified in 4.4, 4.5, 4.6 and 4.7. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQL's) as specified in the contract or order, except where otherwise indicated (see 6.2).

4.3.1 In-process inspection. Visual and dimensional examinations of the fabric and its components, or optional items, shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with requirements specified in section 3. Materials and components that can be classified as a defect in accordance with 4.4.1 through 4.4.1.2 shall be removed from production.

4.4 Inspection methods.

4.4.1 Visual examination. Each roll in the sample shall be examined yard by yard on the face side for defects in accordance with MIL-STD-3064, Type II.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Roll identification and marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for roll identification and marking defects as specified in MIL-STD-3064.

4.4.4 Examination for length. Rolls shall be examined for gross length. When total yardage in the roll exceeds 100 yards, only 100 yards shall be examined (see 6.2). During the yard-by-yard examination, each roll in the sample be examined for shade variation as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for characteristics listed in table II. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 5 continuous yards, full width of finished cloth. The sample size shall be as follows and the lot shall be unacceptable if one (1) or more sample units fail to meet any requirement.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to 22,000	3
22,001 and over	5

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4.5.1 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the samples is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls (see 6.2).

4.6 End item inspection. The cloth shall be tested for the end item characteristics as specified in table II.

Table II. End item inspections.

Characteristic	Requirement Paragraph	Verification Methods
Ball Burst Test	3.3	ASTM D3787
Visual Shade Match	3.3	4.6.1
Weight	3.3	ASTM D3776/D3776M, Option C
MVTR - Inverted	3.3	ASTM E96/E96M
Stretch IP4 % % Elongation Warp % Recovery Warp	3.3	ASTM D2594/D2594M
Color Fastness Crocking Dry Wet	3.3	AATCC TM116 and/or AATCC TM8
Low Pressure Hydrostatic Testing - Initial - After home laundry cycles - After Synthetic Perspiration - After Aircraft Fluid	3.3	AATCC TM127 AATCC TM135, Washing Option (3) (III), Drying (Aiii) AATCC TM15, 1/, 2/ 4.6.3
Color Fastness Perspiration	3.3	AATCC TM15, 3/, 4/
Spray Rating	3.3	AATCC TM22
Dimensional Stability (%)	3.3	AATCC TM135, Washing Option (3) (III), Drying (Aiii)

1/ The perspiration reagent shall be prepared in accordance with AATCC TM15, paragraph 6.
2/ The exposure of the specimen to the contaminant shall be conducted in accordance with AATCC TM15, paragraph 9.1.
3/ The specimen shall be rated using AATCC EP 1.
4/ The specimen shall be rated using AATCC EP 2 against nylon test fabric.

4.6.1 Visual shade matching. The color and appearance of the laminated cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with a primary light source simulating the spectral quality of average daylight,
CIE Illumination D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of

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100 (± 20) foot candles. Alternatively, the color and appearance of the laminated cloth shall match the standard sample with a primary light source simulating artificial daylight, CIE Illumination D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candle in lieu of D65 (see 6.5.1). The laminated cloth shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illumination A, with a color temperature of 2856K (± 200).

4.6.2 Inverted water method (BW). The inverted water method for determining MVTR shall be conducted as specified in ASTM E96/E96M with the test dish in the inverted position (Procedure BW). The free stream air velocity shall be 550 ± 50 FPM as measured 2 inches below the cloth specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall run for 2 hours and weight measurements shall be taken only at the start and completion of the test. The minimum value shall be obtained by averaging all of the determinations obtained from the individual specimens taken from all of the sample units as a group. No single specimen determination shall be less than 3500 grams/meter^2/24 hours.

4.6.3 Contamination procedure JP-8. The specimen (or specimen area) shall be laid flat, face side up, on a glass plate. One milliliter of JP-8 fuel conforming to MIL-DTL-83133 shall be spread over the middle of the sample. The samples shall be placed flat in an aircrew circulating oven at 50 °C for 30 minutes. Remove immediately and test as specified in table II.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in house contractor personnel, these personnel need to contract the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the fabrication of socks for anti-exposure protective clothing.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Application Government including revisions and non-government documents cited under section 2.
c. Type, class, and quantity desired (see 1.2).
d. National stock number.

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e. First article and conformance inspection requirements (see 4.2 and 4.3).
f. Name and address of the first article inspection facility; and the name and address of the Government activity responsible for conducting the first article inspection program.
g. Packaging requirements (see 5.1).

6.3 Information request. For access to information such as standard shade samples of cloth etc. go to Defense Supply Center Philadelphia's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Request to use equivalent materials should be sent to the contracting officer for approval by military services.

6.3.1 Light source. In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.4 Subject term (key word) listing.

Sock
Nylon

CONCLUDING MATERIAL

Custodian:
Navy – AS

Preparing activity:
Navy – AS
(Project 8305-2023-011)

Review activity:
Navy – NU
Air Force – 11
DLA – CT
Army – GL

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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